

Regulatory Decentralization and Construction Productivity

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Two high-level facts about the residential construction sector:

1. Industry:

- Dominated by small, local firms
- Low productivity growth over several decades

2. Regulation:

- Decentralized and managed locally by cities and counties.
- Increased in restrictiveness **and** scope over time

Our Question: Can the decentralized development of municipal building codes explain small firm size and low productivity in the residential construction industry?

- Firm specialization and lost economies of scale
- Distinguish levels from heterogeneity in regulation

Nationwide Analyses

- Historical trends and contemporary estimates of relationship between decentralization and productivity

Data Collection:

1. **California Building Standards Code:** measure levels and heterogeneity in regulations
2. **California Building Permits Data:** link with contractor licenses and study patterns of builder entry into jurisdictions

Preliminary Results

- Cities with that amend their building codes similarly to their neighbors' have higher construction productivity.
- Evidence of large developers clustering in jurisdictions with similar building codes

Work in Progress

- Refining developer name harmonization for mid-size builders.
- Making more detailed measures of similarity in building codes across jurisdictions.

Future Work

- More rigorous identification of productivity losses
- Identify preferences over regulations
- Build a model to understand tradeoff between local preferences and lost productivity

Goal: Welfare calculations of code harmonization, incorporating worse matching of local preferences against increases in construction productivity.

Construction Productivity

- Recently reopened question [Allen, 1985, Syverson and Goolsbee, 2025, Garcia and Molloy, 2025, D'Amico et al., 2024]
- Policy research says lack of standardization hinders manufactured home construction [Maduka et al., 2025] and concentration in SFH construction [Ahluwalia et al., 2022]

Jurisdictional Fragmentation

- Causes and consequences of local political fragmentation [Tiebout, 1956, Epple et al., 1984, Hoxby, 2000, Cassidy and Velayudhan, 2026, Jannin and Sotura, 2020, Bordeu, 2025, Ruggieri, 2025, McCasland et al., 2026]
- Tests fundamental tradeoff of matching local preferences to economies of scale

Text as Data: Measure of Local Building Standards Deviations

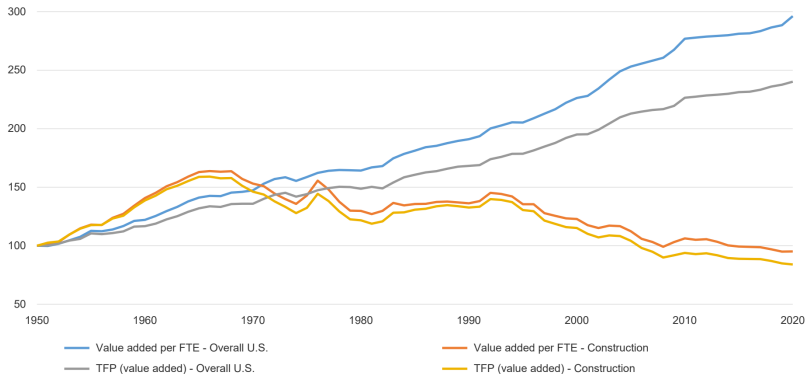
- First dataset built from legal text on building standards: *how* to build, and how much cities *differ*, not only how much regulation they have.

Fact 1: Construction Productivity has Declined

Indexes of Value Added Per Worker and TFP

Overall U.S. and Construction Sector (BEA Data)

Independent reconstruction of Goolsbee and Syverson (2023), Figure 1 | 1950 = 100

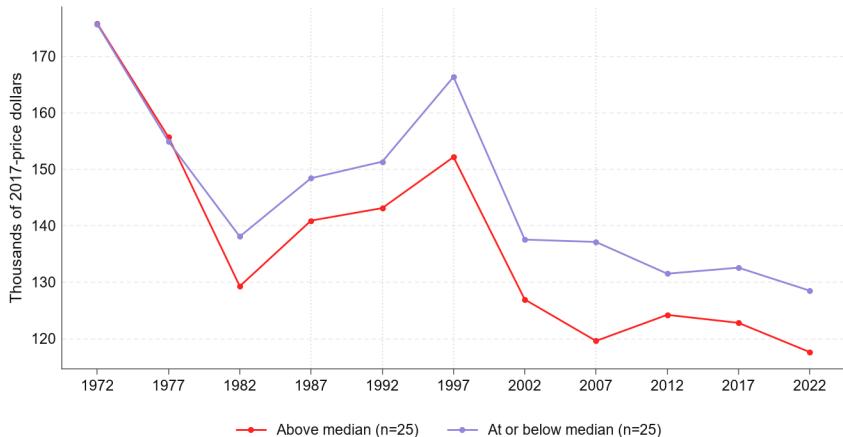


Sources: BEA national, industry, and fixed-assets accounts; Federal Reserve H.15 Treasury-bill rates via FRED.
TFP uses a realized real bill rate and Divisia factor-cost weights. Construction output vintages are linked in 1997.
All four series cover 1950-2020. Data vintages and the explicit interest-rate choice differ from an exact authors-code replication.

Fact 2: Decline is Worse in More Decentralized States

Real value added per employee

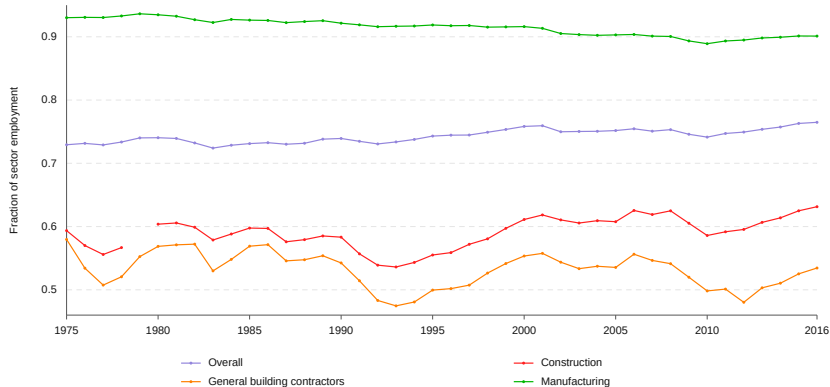
2018 jurisdictions per 100,000 residents | Equal state weights



Fact 3: Construction Firm Sizes are Small

Employment in establishments with 20 or more employees

Partial reconstruction of D'Amico et al. (2024), Figure 4 | Available source data: 1975-2016

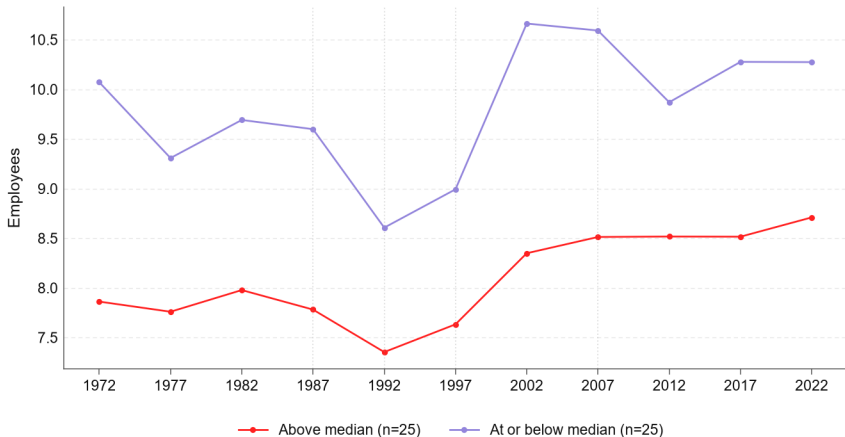


Source: Census County Business Patterns national files; 1975-1985 raw files via Eckert et al.'s archive.
Share = (total employment - employment in establishments with fewer than 20 employees) / total employment.
Public bins identify 20+. The 1979 construction value is suppressed. No graph digitization or imputation.

Fact 4: Firm Sizes are Smaller in More Decentralized States

Employees per establishment

2018 jurisdictions per 100,000 residents | Equal state weights



Question: Do these patterns hold accounting for restrictiveness of regulation and other differences?

Answer: MSA-level analysis of decentralization on construction productivity

- Using 2010-2019 productivity measures from Census Building Permits Survey and County Business Patterns
- Control for population quartic and **WRLURI average**.
- Instrument for jurisdictional counts with 1942 independent school district counts
 - Incorporation and construction productivity both respond to demand.
 - 1942 districts may have influenced 1940-80 wave of municipal incorporation, but were mostly consolidated since.

Reduced Form Analysis

$$\bar{y}_{msa,2010-19} = \alpha + \beta \log \text{Jurisdictions}_{msa} + \gamma \overline{\text{WRLURI}}_{msa} + g(\log(\text{Pop}_{msa,2010})) + \epsilon_{msa}$$

	Log units/ worker	Log units/ establishment	Log workers/ establishment	Log establishments
OLS	-0.073 (0.049)	-0.130** (0.050)	-0.059*** (0.017)	0.026 (0.036)
School districts IV	-0.666*** (0.153)	-0.851*** (0.173)	-0.190*** (0.053)	0.250*** (0.094)
Observations	334	334	334	334
Population quartic			Yes	
WRLURI + missing indicator			Yes	
IV first-stage F			69.22	

Log jurisdictions; HC1 standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

What this says

- Decentralization is associated with fewer units built per worker, smaller establishments, and more establishments
- Robust to controlling for overall level of regulation

What this can't say

- Mechanisms of the relationship
 - Entry costs? Specialization? Quality differences?
- Tradeoff against local regulatory preferences

⇒ **Motivates taking a closer look at developer behavior across jurisdictions!**

Motivation

- Standardized statewide building code with local deviations eases measurement
- Statewide general contractor registries enables matching homebuilders to building permits.
- Large, geographically diverse with multiple population centers.

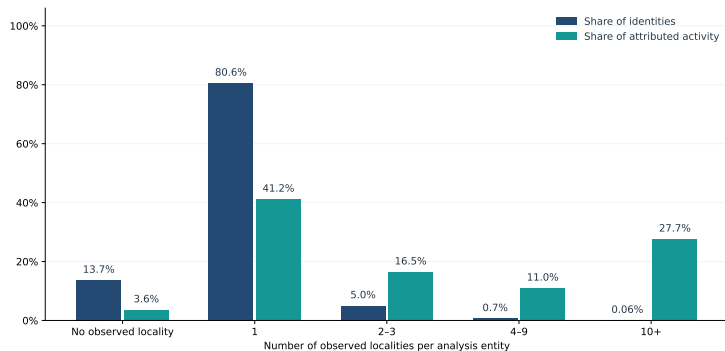
Goals

- Understand geographic patterns of building code deviations
- Document relationship between building codes, developer entry into jurisdictions, and construction output

Developer dataset: cleaning and assembly

1. **Select new construction.** Classify BUILTY permits using text and permit fields; exclude explicit nonresidential types, ADU-only projects, renovations, and standalone trade work.
2. **Link contacts.** Match to CSLB by license, name, phone, or address; retain eligible contacts on permits without registry matches, excluding owner-only roles.
3. **Harmonize businesses.** Standardize names, consolidate exact business matches across licenses, and accept near-matches with supporting contact evidence.
4. **Identify corporate families.** Link businesses to documented subsidiaries from SEC subsidiary exhibits and OpenCorporates following Gorback et al. [2025]

Most firms are local; most activity spans localities



80.6% of firms appear in one locality. firms spanning multiple localities account for **55.2% of activity**

How the California Building Code Is Made



- Title 24: 13 parts (Building, Residential, Fire, CALGreen, ...), republished every **3 years**
- 2022 cycle: published July 1, 2022; in force statewide Jan 1, 2023; local adoption within **180 days**.
- A local amendment must be **filed with the CBSC** to take effect, and **re-filed every cycle**

Two kinds of amendment

- **Building standards:** technical requirements (e.g. concrete strength in a foundation); need an **express finding** of local climatic, geological or topographical conditions; **may not be less restrictive** than Title 24
- **Administrative:** permits, fees, plan sets, inspections; generally no finding needed

The Corpus: Every Local Amendment Filed with the CBSC

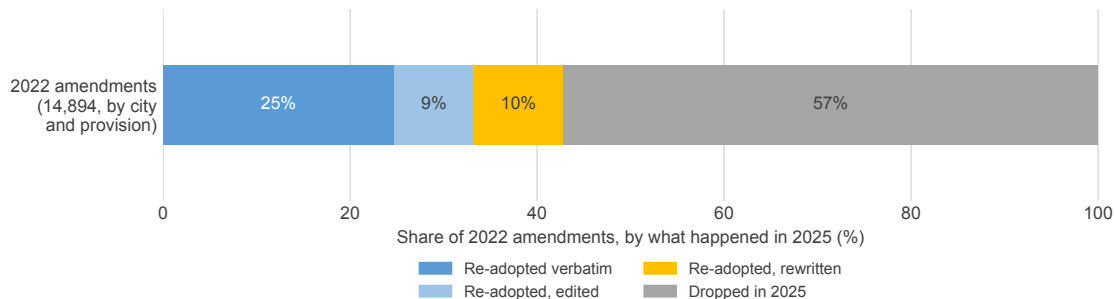
- CBSC *Ordinances Accepted for Filing*
 - **2022 cycle**: 370 filings
 - **2025 cycle**: 358 filings
- **728 PDFs, 50k pages** (median 40 pages, max > 1k)
- Extracted with a large vision model (Qwen3-VL-235B-A22B) [details](#) [extraction](#)
- Unit of observation: **amendment**
 - One row per **ordinance** × **Title 24 section or subsection amended**
 - administrative vs. building standard
 - justification (linked to findings)
 - verbatim amended text
 - adoption and effective dates

- **46k amendments** in 728 filings
 - 2022 cycle: 25k (288 jurisdictions)
 - 2025 cycle: 21k (236 jurisdictions)
- Jurisdictions amending Title 24: 52% of the 483 cities in 2022 (45% in 2025 cycle); 47% of 58 counties (21 in 2025 cycle).
- *2022 cycle (288 jurisdictions, 370 filings):*
 - **75%** building standards, **24%** administrative, 1% unclassified
 - Median jurisdiction: **67 amendments** (mean 88)
 - Across all of Title 24, local jurisdictions changed **2,063 sections**:
 - most changes are idiosyncratic: **56%** of sections are changed by only one jurisdiction
 - In common: **265** sections are changed by 10 or more jurisdictions, and 27 by 50 or more (e.g. sprinklers, permits, roofing)

Example: Cupertino

Timeline 2022

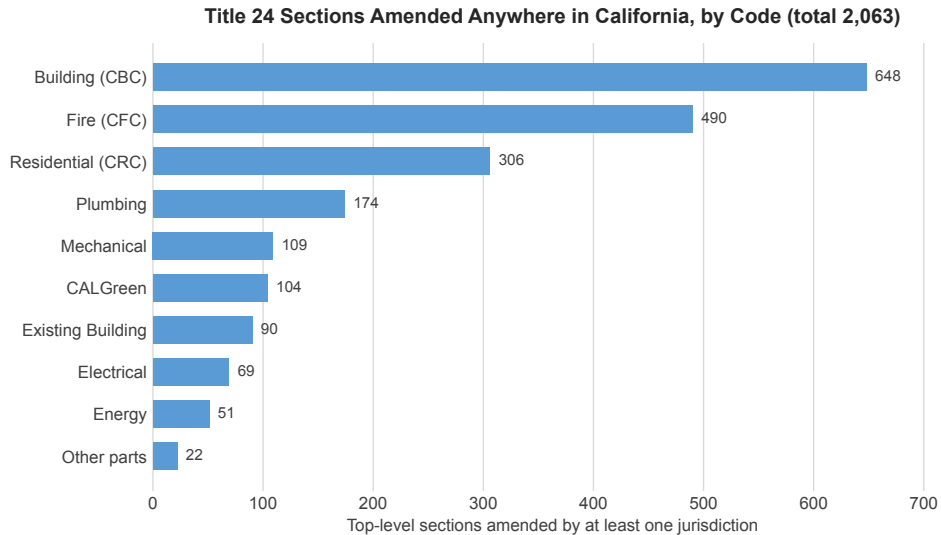
Re-adoption: How Much Carries Over Between Cycles?



188 jurisdictions that amended in both cycles; 14,894 amendments in 2022 (city x provision). Re-adopted = same Title 24 section amended in both cycles. Verbatim: cosine similarity of TF-IDF word vectors ≥ 0.95 ; edited 0.80-0.95; rewritten < 0.80 . Findings-resolution records excluded.

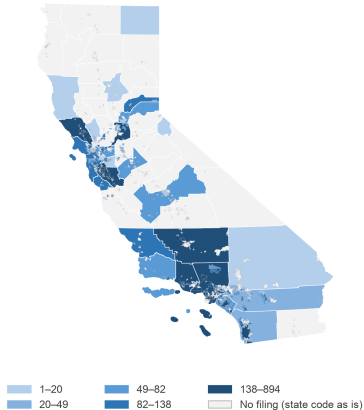
- Two-thirds of 2022 amenders amend again in 2025 (188 of 287)
- Only **a quarter** of 2022 amendments are re-adopted verbatim in 2025; **57%** are dropped

Which Codes Do Cities Amend?



2022 Cycle: Where Amendments Happen; All Amendments

All Local Amendments



Source: local amendment ordinances filed with the California Building Standards Commission, 2022 code cycle. Cities drawn over their county; unincorporated areas take the county's filing. Census TIGER 2025 boundaries.

- Amendments in the 2022 cycle, by jurisdiction
- All cities drawn over their county; county color = unincorporated area only
- Grey: no filing, i.e. Title 24 adopted as is (about half of all cities)
- Most active:
 - Glendale (894)
 - West Hollywood (726)
 - Kern County (408)
 - Sacramento County (360)
 - Malibu (350)
 - Beverly Hills (337)

2025 Outcomes Under the 2022 Code: Amendment Counts

	Units / 1k pop.	SF units / 1k pop.	MF units / 1k pop.	Units per job
Amendments (per 100)	0.26*** (0.06)	-0.01 (0.10)	0.43*** (0.15)	0.28*** (0.07)
Administrative Amendments (per 100)	0.46 (0.28)	0.06 (0.35)	1.66** (0.66)	0.47 (0.30)
Building-Standard Amendments (per 100)	0.32*** (0.07)	-0.02 (0.10)	0.45** (0.19)	0.35*** (0.09)
Filed indicator	✓	✓	✓	✓
log population	✓	✓	✓	✓
City geography	✓	✓	✓	✓
County FE	✓	✓	✓	✓
Cities	373	366	165	333

OLS of log 2025 outcome; each row and column is a separate regression. Regressor: the city's 2022-cycle amendments that amend, add or replace a Title 24 provision (per 100; deletions excluded). Cities with zero outcome dropped; non-filing cities enter as 0. City geography: latitude, log land area and log distance to the coast. S.e. clustered by county in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

without county FE

Three Measures of How Different Neighbors' Codes Are

- A jurisdiction's code: the set of Title 24 sections it amended (A, B); state-code city = $\emptyset$
- Computed for each land-bordering neighbor, then averaged over the city's neighbors

		Saratoga – Los Gatos 52 and 64 sections, 31 shared	Albany – Berkeley 0 and 65 sections	Two state-code cities
1. Similarity	$1 - \frac{ A \triangle B }{2,063}$: share of contested sections that agree	$1 - \frac{54}{2063} = \mathbf{97.4\%}$	96.8%	100%
2. Jaccard (amenders)	$\frac{ A \cap B }{ A \cup B }$: shared / either amended	$\frac{31}{85} = \mathbf{36.5\%}$	not defined	not defined

- **Similarity:** neighbors agree on **98%** of the 2,063 sections amended anywhere in California
- **Jaccard (amenders):** when both neighbors amend, they rarely pick the same sections: the median city shares **9%** with their neighbors.

Details measures

Similarity With Neighbors: Bay Area vs. Los Angeles County

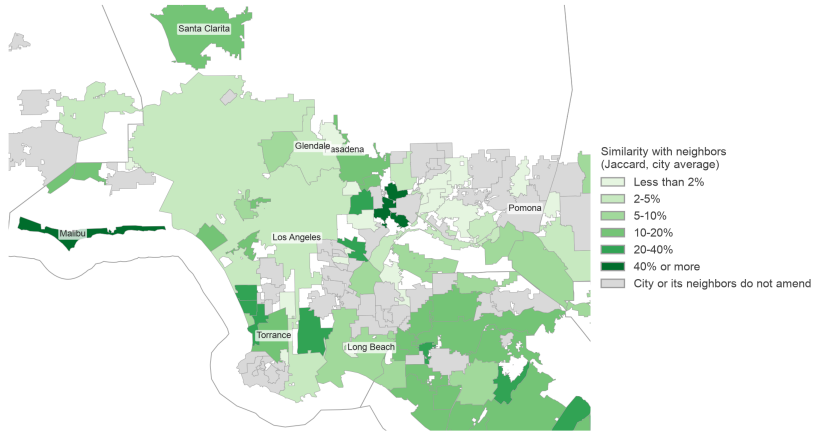
	Bay Area	Los Angeles County
<i>All cities</i>		
Share of cities that amend	75%	57%
Sections that differ from a neighbor, mean	52	85
Similarity with neighbors, mean	97.5%	95.9%
<i>Cities that amend, with an amending neighbor</i>		
Sections the city amended, median	32	52
Jaccard similarity, mean	11.4%	12.6%
median	8.8%	8.4%

- More Bay Area cities amend, but LA cities that amend change more sections $\Rightarrow$ LA neighbors differ on more rules (85 vs. 52)

LA

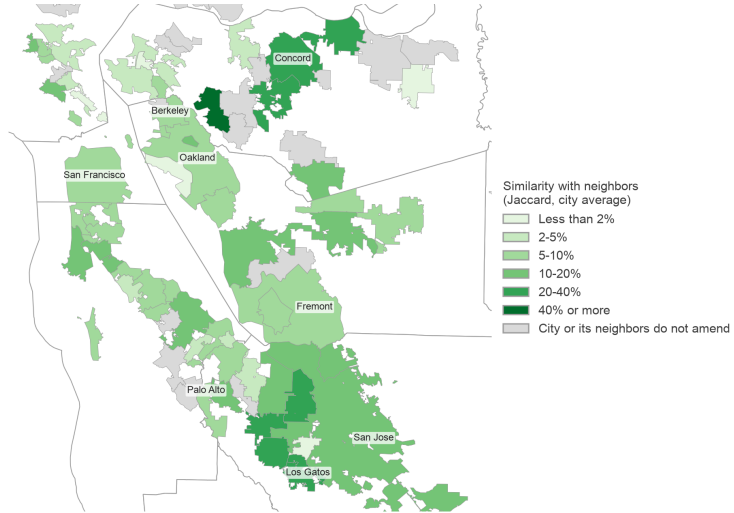
Bay Area

Jaccard Similarity Among Amenders: Los Angeles County



Jaccard similarity: top-level sections both amended / sections either amended, for each amending jurisdiction sharing a land border with the city (cities, and the county government where the city borders unincorporated land), averaged. 2022 cycle. White: unincorporated land.

Jaccard Similarity Among Amenders: Bay Area



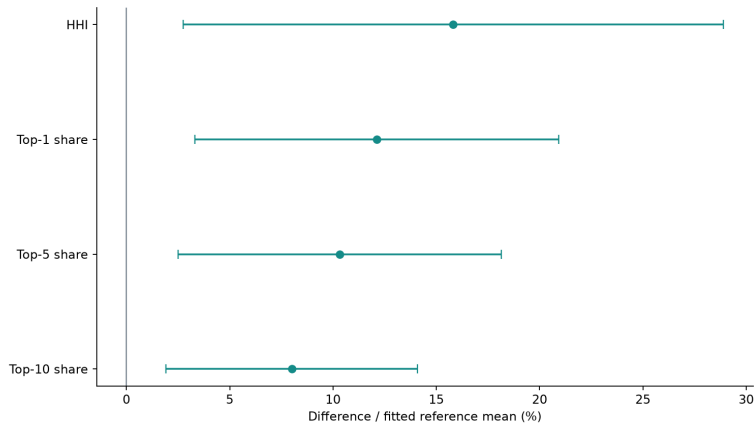
Jaccard similarity: top-level sections both amended / sections either amended, for each amending jurisdiction sharing a land border

Result 1: Coordination with neighbors is associated with higher construction productivity

	Units / 1k pop.	SF units / 1k pop.	MF units / 1k pop.	Units per job
<i>Panel A: All cities</i>				
1. Similarity (per pp)	-0.031	0.065*	-0.088***	-0.001
	(0.041)	(0.035)	(0.030)	(0.029)
Cities	426	419	196	377
<i>Panel B: Cities that amend, with an amending neighbor</i>				
2. Jaccard similarity (per 10 pp)	0.201**	0.196***	0.239	0.263***
	(0.078)	(0.071)	(0.188)	(0.099)
Cities	212	209	110	204
log population	✓	✓	✓	✓
City geography	✓	✓	✓	✓
County FE	✓	✓	✓	✓

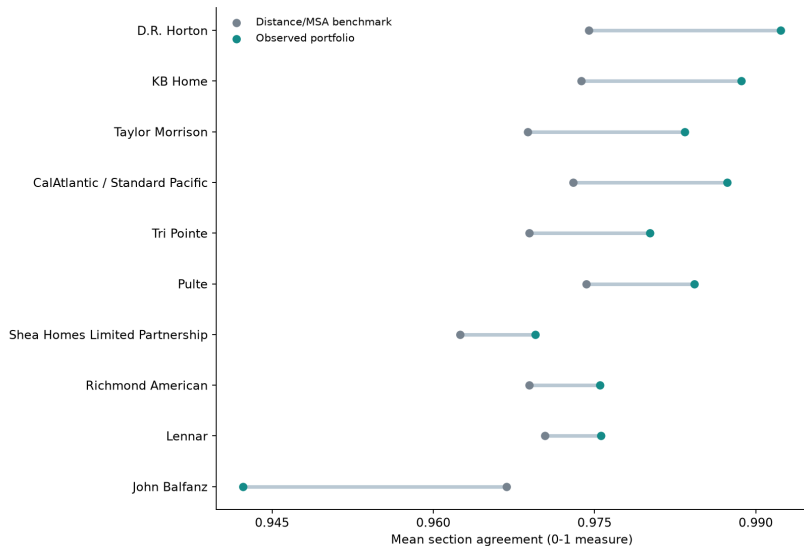
Note: OLS of log outcome; each row and column is a separate regression. 2022 cycle, top-level Title 24 sections, measures averaged over land-bordering neighbors (cities, and the county government where the city borders unincorporated land). Controls: log population, city geography (log latitude, log land area, log distance to the coast) and county fixed effects. S.e. clustered by county. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Result 2: Neighbor similarity is associated with higher developer concentration



Controls: record count; no county fixed effects. Cities with ≥ 30 records.
County-clustered 95% delta-method intervals. Reference is the fitted outcome at median exposure.
Percentages are not percentage points; Jaccard excludes nonamenders.

Result 3: Large builders cluster in jurisdictions with similar codes



We ask: how does regulatory decentralization effect construction productivity?

We answer:

- In a nationwide analysis, more decentralized markets have smaller firm sizes and lower output per worker.
- In California, building code similarity is associated with higher firm concentration and higher output per worker, with large developers clustering in similar jurisdictions.

We Contribute:

- New questions and answers on construction productivity and regulatory decentralization
- New insights and data on building codes in California

Thank you!

Example: Cupertino Amends CBC 903.2 (Where Sprinklers Are Required)

State code (2022 CBC 903.2)

- Sprinklers required by **occupancy and size**, section by section (903.2.1–903.2.12)
- Many commercial uses only above a fire-area threshold (e.g. **12,000 sq ft** for retail and storage)
- Homes: sprinklers already required statewide (CRC R313)

Cupertino (Ord. 22-2245, re-adopted as Ord. 25-2278)

- Sprinklers throughout **all new buildings**, except:
 - up to **1,200 sq ft** (**500 sq ft** in wildland-urban interface areas)
 - open parking structures, fuel canopies, temporary trailers
- All new buildings with a residential (Group R) fire area
- Existing homes: sprinklers when additions push the fire area above **3,600 sq ft**
- New manufactured homes

Appendix: Developer dataset inputs

- BUILTY permits** 30.82 million California source rows: permit descriptions, work and property types, addresses, dates, valuations, and project contacts.
- California APR** Table A2 housing records: addresses, parcel numbers, unit categories, and permit/occupancy dates. Supplies the non-ADU housing backbone.
- CSLB registry** Contractor license numbers, business names, license classes, and contact details. Supplies business identity and linkage evidence.
- Corporate records** SEC subsidiary exhibits and OpenCorporates relationship statements. Supply parent–subsidiary evidence.
- Census geography** Census Geocoder and TIGER/Line boundaries. Assign construction addresses to municipalities, counties, and MSAs.

Appendix: The Model

Qwen3-VL-235B-A22B (open-weight)

- Vision-language model: reads each page **as an image**
- Output forced to our **JSON schema** (structured decoding), temperature 0
- Before extracting, lists every section number visible on the pages, then one record per provision

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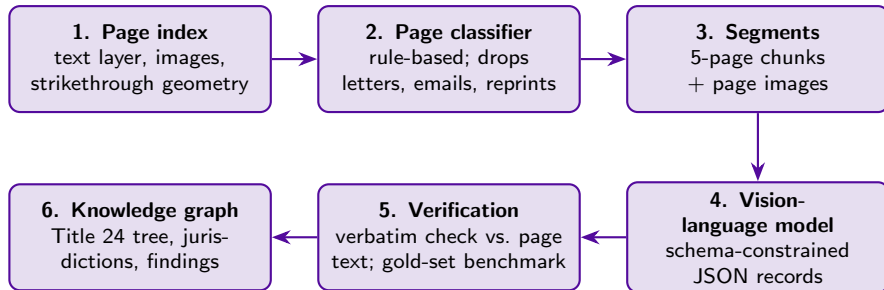
Benchmark: Claude Opus

- Same schema, same prompts
- Run on a stratified **40-filing gold set**
- Used to score Qwen field by field.

gold set

- Open weights: the exact model version is fixed and public $\Rightarrow$ the extraction is reproducible

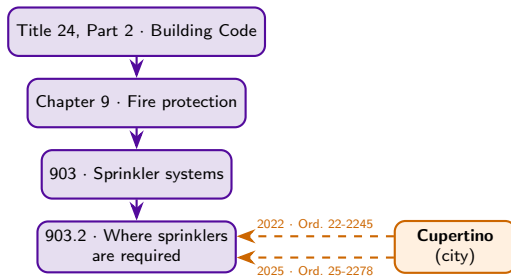
Appendix: From PDFs to a Knowledge Graph



Knowledge Graph

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Appendix: From Filings to a Knowledge Graph



Purple: state code (section tree)

Orange: jurisdiction; one dashed edge per amendment, carrying its text, type, ordinance and date

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Appendix: Validation Against the Gold Set

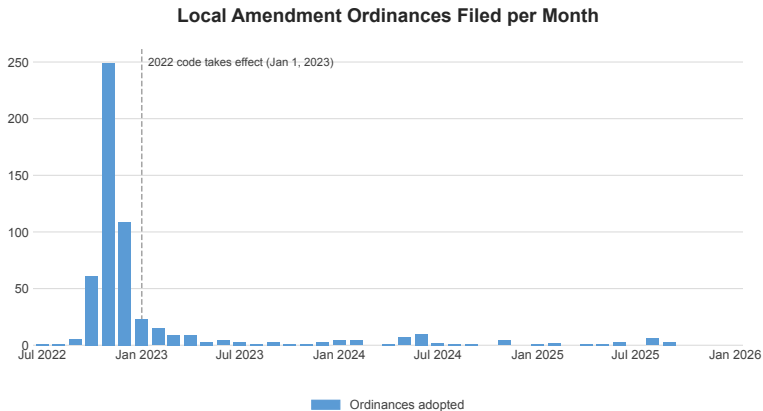
- 40 filings, stratified
- Extracted with Claude Opus using the same schema; Qwen scored against it

Measure	Qwen vs. gold
Sections found (share of gold sections recovered)	93%
Operation (amend / add / delete) agreement	88%
Administrative vs. building standard agreement	77%
Title 24 part agreement (after rule-based resolution)	70%

Qwen rarely answers “unclear” on administrative vs. building standard, so its building-standard share is an upper bound.

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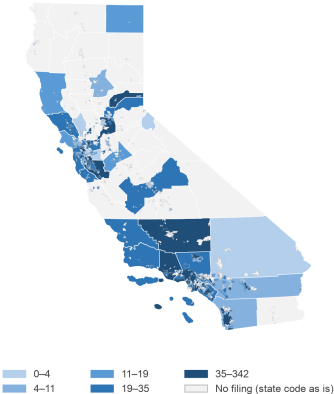
Timeline of Local Amendment Ordinances (2022 Cycle)



Source: local amendment ordinances filed with the California Building Standards Commission, 2022 code cycle (370 filings).
Amendments that change or delete a Title 24 section; adoptions by reference, appendix adoptions and municipal-code-only provisions excluded.

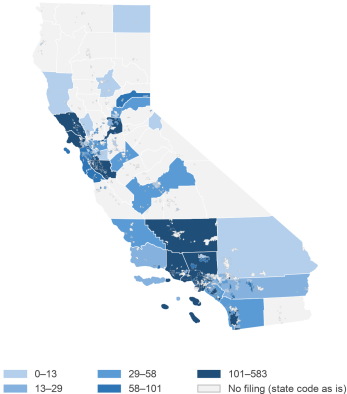
Appendix: Administrative vs. Building-Standard Amendments (2022 Cycle)

Administrative only



Source: local amendment ordinances filed with the California Building Standards Commission, 2022 code cycle.
Cities drawn over their county; unincorporated areas take the county's filing. Census TIGER 2025 boundaries.

Building standards only



Source: local amendment ordinances filed with the California Building Standards Commission, 2022 code cycle.
Cities drawn over their county; unincorporated areas take the county's filing. Census TIGER 2025 boundaries.

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When Cities Change the Code, They Change It Differently

	Mean	Median
<i>All cities</i> (482; state-code cities included)		
Similarity with neighbors (share of contested sections that agree)	98%	98%
Sections that differ from a neighbor	43	33
Cities identical to all their neighbors		16%
<i>Cities that amend, with an amending neighbor</i> (228)		
Sections the city amended	39	29
Jaccard similarity: shared / either amended	12%	9%
Cities sharing no amended section with any neighbor		9%
Cities below 5% Jaccard		29%

- Neighbors' codes agree on most rules: most of Title 24 is left at the state text
- But amendments rarely line up: of the sections two neighbors changed, the median city shares **9%**

Appendix: Similarity With Neighbors, Bay Area Cities

Most similar					Least similar				
City	County	Sections	Nbrs.	Jaccard	City	County	Sections	Nbrs.	Jaccard
Orinda	Contra Costa	20	1	56.7%	Newark	Alameda	17	1	5.4%
Concord	Contra Costa	39	3	39.5%	San Mateo	San Mateo	32	5	5.3%
Monte Sereno	Santa Clara	55	3	38.1%	Sausalito	Marin	29	1	5.1%
Walnut Creek	Contra Costa	39	2	33.6%	Sonoma	Sonoma	17	1	4.0%
Santa Rosa	Sonoma	49	1	33.3%	Hillsborough	San Mateo	15	3	3.8%
Healdsburg	Sonoma	45	1	28.2%	San Rafael	Marin	11	3	3.1%
Pittsburg	Contra Costa	39	2	28.0%	Tiburon	Marin	11	3	0.5%
Los Gatos	Santa Clara	62	5	26.0%	St. Helena	Napa	23	1	0.0%
Saratoga	Santa Clara	52	6	25.0%	Napa	Napa	10	1	0.0%
Cupertino	Santa Clara	86	5	23.4%	Cotati	Sonoma	10	2	0.0%
All 74 cities: mean Jaccard 11.4%, median 8.8%									

Sections: distinct Title 24 sections the city amended (2022 cycle, text changes). Nbrs.: land-bordering jurisdictions that amend, including the county government where the city borders unincorporated land. Similarity: average Jaccard similarity with those neighbors. County mean in parentheses. Cities that amend and have at least one amending neighbor.

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Appendix: Similarity With Neighbors, Los Angeles County Cities

Most similar				Least similar			
City	Sections	Nbrs.	Jaccard	City	Sections	Nbrs.	Jaccard
South El Monte	132	2	72.8%	Arcadia	35	4	3.0%
Malibu	139	1	64.6%	Walnut	17	3	2.9%
Rosemead	122	5	62.9%	Pico Rivera	28	6	2.8%
Temple City	115	3	50.0%	Monterey Park	26	4	1.9%
Manhattan Beach	51	4	29.5%	Hidden Hills	15	3	1.9%
Carson	123	3	29.3%	Gardena	27	3	1.5%
Redondo Beach	53	4	23.0%	Baldwin Park	12	4	0.9%
El Segundo	72	3	23.0%	Santa Fe Springs	10	4	0.4%
Commerce	121	5	20.8%	Lancaster	21	2	0.0%
Alhambra	122	5	20.1%	West Covina	19	5	0.0%

All 50 cities: mean Jaccard 12.6%, median 8.4%

Sections: distinct Title 24 sections the city amended (2022 cycle, text changes). Nbrs.: land-bordering jurisdictions that amend, including Los Angeles County where the city borders unincorporated land. Similarity: average Jaccard similarity with those neighbors. Cities that amend and have at least one amending neighbor.

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Appendix: Geography and Hazard Controls

- **Location:** Census internal point of the place or county (TIGER 2025)
- **Elevation and ruggedness:** mean and s.d. of USGS elevation at nine points inside the jurisdiction
- **Distance to the ocean:** internal point to the TIGER coastline
- **Hazards:** FEMA National Risk Index v1.20, county level; expected annual building-loss rate by hazard, in logs (*risk scores* are national percentiles)
- Cities take their county's hazard values; special districts (no Census geometry) are excluded

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Appendix: Known Limitations

- Building vs. Fire Code attribution is approximate (5-page chunks often miss the heading naming the code)
- 7 of 9,269 chunks not extracted (scanned pages where the model loops)
- 23 filings with no amendments: 9 lost at page classification, 14 to check by hand (e.g. Port of San Francisco)
- CBSC deficiency tables not yet re-extracted for the full corpus
- Adoption dates stated for 73% of ordinances

Result 1: Similarity Is Associated With Higher Construction Productivity

	Units / 1k pop.	SF units / 1k pop.	MF units / 1k pop.	Units per job
<i>Cities that amend, with an amending neighbor</i>				
1. Similarity (per pp)	-0.017 (0.025)	0.089*** (0.016)	-0.126*** (0.042)	-0.023 (0.030)
2. Jaccard similarity (per 10 pp)	0.201** (0.078)	0.196*** (0.071)	0.239 (0.188)	0.263*** (0.099)
Cities	212	209	110	204
log population	✓	✓	✓	✓
City geography	✓	✓	✓	✓
County FE	✓	✓	✓	✓

Note: OLS of log outcome; each row and column is a separate regression. 2022 cycle, top-level Title 24 sections, measures averaged over land-bordering neighbors (cities, and the county government where the city borders unincorporated land). Controls: log population, city geography (log latitude, log land area, log distance to the coast) and county fixed effects. S.e. clustered by county. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

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Neighbors' Codes and Construction Outcomes: Building-Standard Amendments

	Units / 1k pop.	SF units / 1k pop.	MF units / 1k pop.	Units per job
<i>Panel A: All cities</i>				
1. Similarity (per pp)	-0.038 (0.044)	0.076** (0.036)	-0.088** (0.044)	-0.005 (0.033)
Cities	426	419	196	377
<i>Panel B: Cities that amend, with an amending neighbor</i>				
2. Jaccard similarity (per 10 pp)	0.165** (0.072)	0.163** (0.065)	0.196 (0.131)	0.248*** (0.086)
Cities	210	207	110	202
log population	✓	✓	✓	✓
City geography	✓	✓	✓	✓
County FE	✓	✓	✓	✓

Note: Panel A: all California cities; Only includes building standard amendments. Controls: log population, city geography (log latitude, log land area, log distance to the coast) and county fixed effects. S.e. clustered by county. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

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Neighbors' Codes and Construction Outcomes: Administrative Amendments

	Units / 1k pop.	SF units / 1k pop.	MF units / 1k pop.	Units per job
<i>Panel A: All cities</i>				
1. Similarity (per pp)	-0.016 (0.037)	0.044 (0.034)	-0.078*** (0.020)	0.004 (0.027)
Cities	426	419	196	377
<i>Panel B: Cities that amend, with an amending neighbor</i>				
2. Jaccard similarity (per 10 pp)	0.284** (0.136)	0.278** (0.131)	0.509* (0.282)	0.246 (0.156)
Cities	194	192	102	187
log population	✓	✓	✓	✓
City geography	✓	✓	✓	✓
County FE	✓	✓	✓	✓

Note: As in the previous table, counting administrative amendments only (permits, fees, inspections, appeals); each row and column is a separate regression.

Controls: log population, city geography (log latitude, log land area, log distance to the coast) and county fixed effects. S.e. clustered by county. * $p < 0.1$,

** $p < 0.05$, *** $p < 0.01$.

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Appendix: 2025 Outcomes Under the 2022 Code, Without County Fixed Effects

	Units / 1k pop.	SF units / 1k pop.	MF units / 1k pop.	Units per job
Amendments (per 100)	0.17*** (0.06)	-0.05 (0.07)	0.33*** (0.12)	0.20*** (0.07)
Administrative Amendments (per 100)	0.14 (0.28)	-0.02 (0.28)	1.04** (0.51)	0.22 (0.35)
Building-Standard Amendments (per 100)	0.22*** (0.07)	-0.06 (0.08)	0.39*** (0.15)	0.26*** (0.08)
Filed indicator	✓	✓	✓	✓
log population	✓	✓	✓	✓
City geography	✓	✓	✓	✓
Cities	373	366	165	333

As in the main table, without county fixed effects. OLS of log 2025 outcome; each row and column is a separate regression. Regressor: the city's 2022-cycle amendments that amend, add or replace a Title 24 provision (per 100; deletions excluded). Cities with zero outcome dropped; non-filing cities enter as 0. Controls: filed indicator, log population, city geography (latitude, log land area, log distance to the coast). S.e. clustered by county. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

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